Photomultiplier XP3312

8-stage
76mm (3"), round tube



Application
✓ Gamma-cameras

Feature
✓ High PHR

Description	
Window material	Lime glass
Photocathode	Bi-alkali
Refr. Index at 420nm	1.54
Multiplier structure	Linear focused

Photocathode characteristics	Min	Typ	Max	Unit
Spectral range :		290-650		nm
Maximum sensitivity at :		420		nm
Sensitivity :				
Luminous :		100		µA/lm
Blue * :	11	13		µA/lmf
Radiant, at 420nm		100		mA/W
Characteristics with voltage divider A	**Min**	**Typ**	**Max**	**Unit**
Gain slope (vs supp. Volt., log/log)		5.8		
For an anode blue sensitivity of		3		A/lmf
Supply voltage *	800	1000	1200	V
Gain		$2.3x10^5$		
Anode dark current *		1	20	
Pulse height resolution ^{57}Co – NaI(Tl) 3"X3"		8.6	9.2	%
Pulse height resolution ^{137}Cs		7		%
Mean anode sensitivity deviation :				
Long term (16h) :		1		%
After change of count rate :		1		%
Under 511 KeV first step operation:		-2		%
Vs temperature between 0 and +40℃ at 420 nm		-0.3		%/K
For a supply voltage of : 1000V	**Min**	**Typ**	**Max**	**Unit**
Linearity (2%) of anode current up to :		10		mA
Anode pulse:				
Rise time:		5		ns
Duration at half height:		11		ns
Transit time:		45		ns

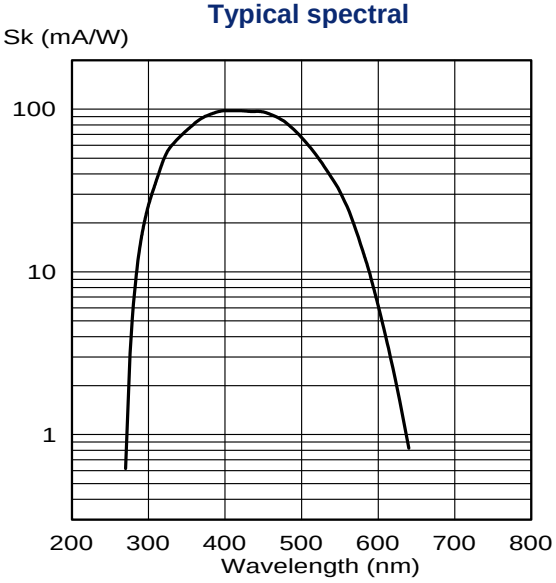
Recommended Voltage Divider

Type A for maximum gain

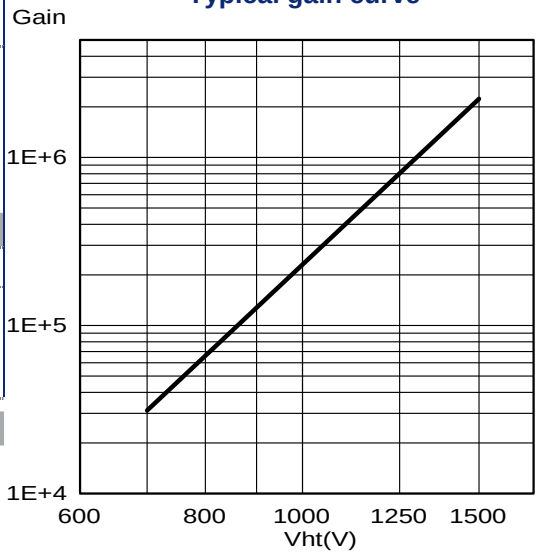
K	D1	D2	D3	D4	D5	D6	D7	D8	A	
2	1,5	1,5	1,5	1	1	1	1	0,5		(total : 11)

**** characteristic mentioned on the test ticket of the tube***

Typical spectral



Typical gain curve

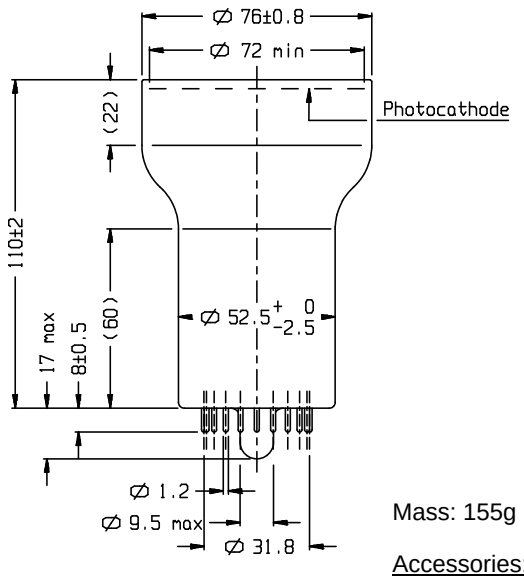


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Photomultiplier XP3312

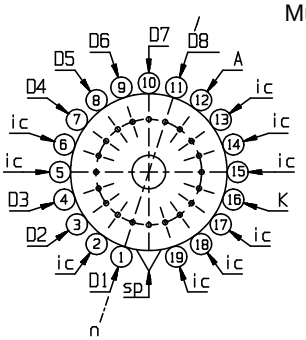
Outline (dimensions in mm)

XP3312

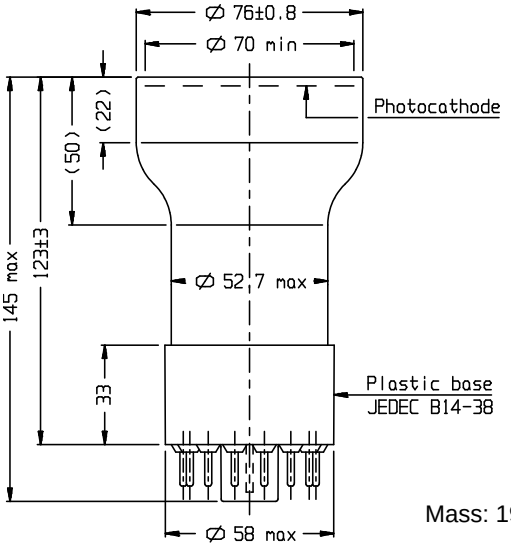


Mass: 155g

Accessories:
Socket: FE2019
Mu-metal shield: MS153

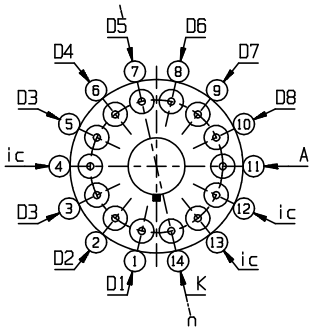


XP3312B



Mass: 195g

Accessories:
Socket: FE1014
Voltage divider: VD282K
Mu-metal shield: MS153



K: cathode
Dn: dynode

A: anode
ic: internal connection

sp: short pin

n: plane of symmetry of the multiplier

Limiting values	Min	Max	Unit
Anode blue sensitivity		30	A/lmf
Supply voltage		1500	V
Continuous anode current		0.2	mA
Voltage between :			
D1 and photocathode :	150	500	V
Consecutive dynode :		300	V
Anode and D8 :	40	300	V
Ambient temperature :			
Short operation (<30 mn) :	-30	+80	℃
Continuous operation & storage :	-30	+50	℃

Variants

XP3312 ▬ ▬

Finishing
B with plastic base JEDEC B14-38
F with flying leads ∅ 0.5
FB with flying leads and plastic base

Option
C with electrostatic coating
(conductive paint connected to the cathode
+ insulating coating)

Also, other variants can be made. Please, contact us to discuss any specific product requirements.

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